Convex Sets

Convex Optimization, Homework 2

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This homework is about recognising and proving convexity of sets. Every answer is a short proof or a counterexample.

Problems and what I did

1. **Affine sets as solution sets of linear equations.** For an affine set $L = x_0 + V$ with V a linear subspace, I wrote V as the null space of a matrix P ($V = \{v : Pv = 0\}$) and showed that $L = \{x : Px = Px_0\}$. So every affine set is the solution set of a system $Ax = b$ with $A = P$, $b = Px_0$.

2. **Which sets are convex?** A slab, a rectangle and a wedge are convex because each is an intersection of half-spaces. The set of points closer to one point than to a given set was written as an intersection of half-spaces. The set $\{x : x + S_2 \subseteq S_1\}$ is convex as an intersection of the convex sets $S_1 - y$. A counterexample in $\mathbb{R}$ ($S = \{-1, 1\}$, $T = \{0\}$) shows that the set of points closer to S than to T need not be convex. The set of points closer to a than to b, scaled by θ, is a Euclidean ball.

3. **Expanded and restricted sets.** For a convex S, both $S_a = \{x : \text{dist}(x, S) \leq a\}$ and S_{-a} (points whose a-ball lies in S) are convex. Both proofs take two points and check their convex combination.

4. **Sets of probability distributions.** For a discrete random variable with distribution p, I checked which constraints on p give convex sets. Bounds on probabilities and expectations, and bounds on $\mathbb{E}f(x)$, are linear in p, so they give convex sets. The variance constraint $\text{var}(x) \geq \alpha$ is not convex; I gave a counterexample with $n = 2$. Quantile constraints reduce to a single linear inequality in p.

5. **Projection onto a compact convex set.** The distance function $y \mapsto \|x - y\|$ is continuous, so it attains a minimum on a compact set C. If two points attained it, their midpoint would also be in C (by convexity), and the triangle inequality forces the two points to coincide. So the closest point is unique.

6. **The ℓ_∞ ball as a polyhedron.** I described the unit ℓ_∞ ball by $2n$ linear inequalities $\pm e_i^T x \leq 1$, and as the convex hull of the 2^n vectors with entries ± 1.